

NATIONAL INSTITUTE OF ENGINEERING

END SEMESTER EXAMINATION - MAY 2027

Duration: 3 Hours

Max Marks: 80

Instructions: Attempt any four questions. Figures to the right indicate full marks.

Q1(a) Explain the layered reference model and justify each layer with a diagram. [4]

Q1(b) Describe the scheduling policy and derive its average waiting time. [4]

Q1(c) Differentiate between the two normalisation forms with a worked example. [4]

Q1(d) Discuss the error control mechanism and analyse its overhead. [4]

Q1(e) Derive the expression for throughput and state all assumptions clearly. [4]

Q1(f) Compare the two indexing structures and evaluate their lookup complexity. [4]

Q1(g) Explain the deadlock avoidance algorithm and apply it to a sample matrix. [4]

Q1(h) Illustrate the encoding scheme and compute the resulting bandwidth. [4]

Q1(i) Analyse the concurrency protocol and prove its serialisability. [4]

Q1(j) Describe the routing metric and calculate the shortest path for the graph. [4]

Q2(a) Discuss the error control mechanism and analyse its overhead. [6]

Q2(b) Derive the expression for throughput and state all assumptions clearly. [6]

Q2(c) Compare the two indexing structures and evaluate their lookup complexity. [6]

Q2(d) Explain the deadlock avoidance algorithm and apply it to a sample matrix. [6]